PRO MINDS

Algebra 1 — Arithmetic Sequences & Scatter Plots

Sequences: common differences and formulas · Scatter plots: association, trend lines, and predictions · 18 questions

Name: _____ Date: _____

Free practice worksheet — show your work in the space under each question, then check yourself with the answer key.

Question 1 · Arithmetic sequences · Is it arithmetic?

6, 11, 16, 21, …

Tell whether the sequence is an arithmetic sequence. If it is, give the common difference.

Question 2 · Arithmetic sequences · Is it arithmetic?

−3, 2, 6, 9, …

Tell whether the sequence is an arithmetic sequence. If it is, give the common difference.

Question 3 · Arithmetic sequences · Is it arithmetic?

30, 24, 18, 12, ⋯

Tell whether the sequence is an arithmetic sequence.
If it is, give the common difference.

Question 4 · Arithmetic sequences · Recursive & explicit formulas

4, 11, 18, 25, ⋯

Write a recursive formula for the arithmetic sequence.

Write an explicit formula for the arithmetic sequence.

Question 5 · Arithmetic sequences · Recursive & explicit formulas

2.5, 3.25, 4, 4.75, ...

Write a recursive formula for the arithmetic sequence.

Write an explicit formula for the arithmetic sequence.

Question 6 · Arithmetic sequences · Recursive & explicit formulas

12, 5, −2, −9, ...

Write a recursive formula for the arithmetic sequence.

Write an explicit formula for the arithmetic sequence.

Question 7 · Arithmetic sequences · Converting formulas

$a_1 = 7,\ a_n = a_{n-1} + 4$

Write an explicit formula for the arithmetic sequence given by this recursive formula.

Question 8 · Arithmetic sequences · Converting formulas

$a_1 = -4,\ a_n = a_{n-1} - 6$

Write an explicit formula for the arithmetic sequence given by this recursive formula.

Question 9 · Arithmetic sequences · Converting formulas

$a_n = 8 + 5n$

Write a recursive formula for the arithmetic sequence given by this explicit formula.

Question 10 · Arithmetic sequences · Reasoning

You are given the first four terms of an arithmetic sequence. Under what conditions might a recursive formula be preferred over the explicit formula? Under what conditions might an explicit formula be preferred over the recursive formula? Explain your reasoning.

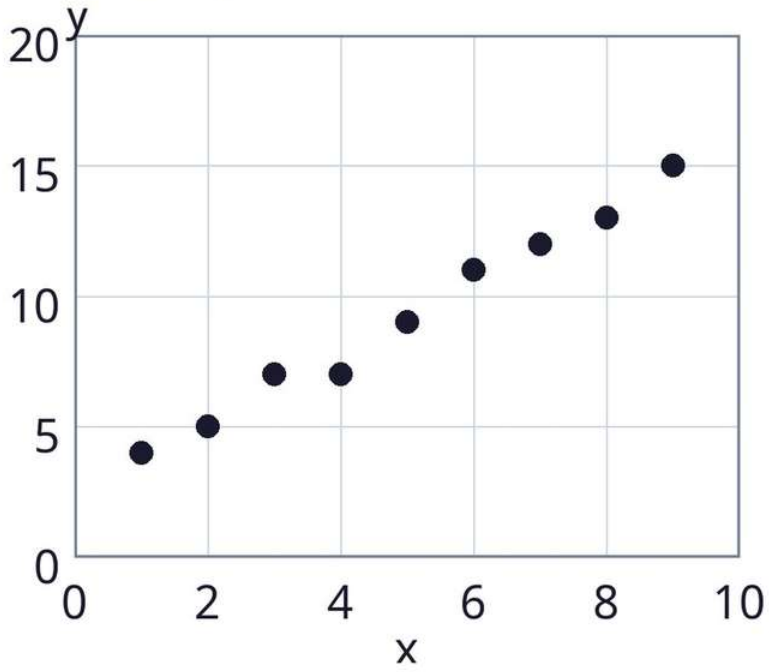
Question 11 · Arithmetic sequences · Application

You open a savings account with a $250 deposit. Each month after that, you deposit $40 into the account.

Write an explicit rule to represent the amount of money in your account after n months. How much money will you have in the account on month 12?

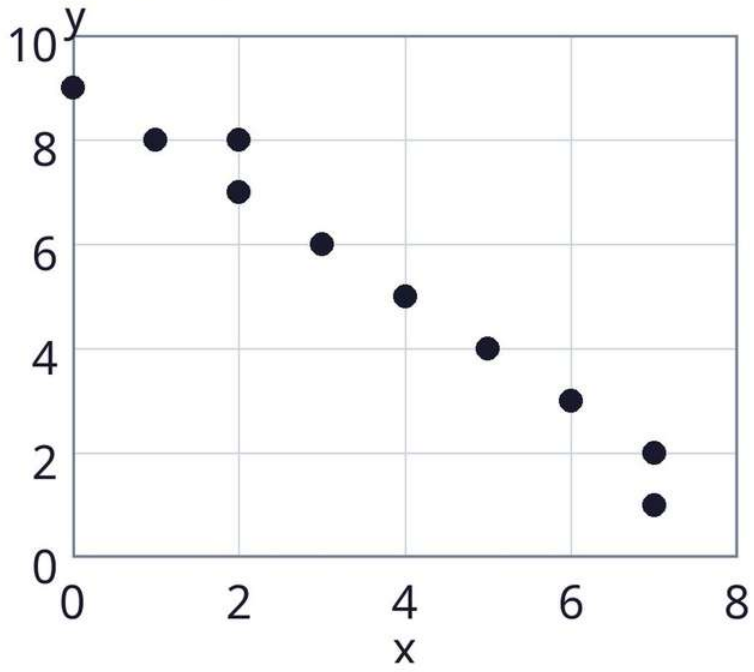
Question 12 · Scatter plots · Association

What is the association between the x- and y-values for this graph?



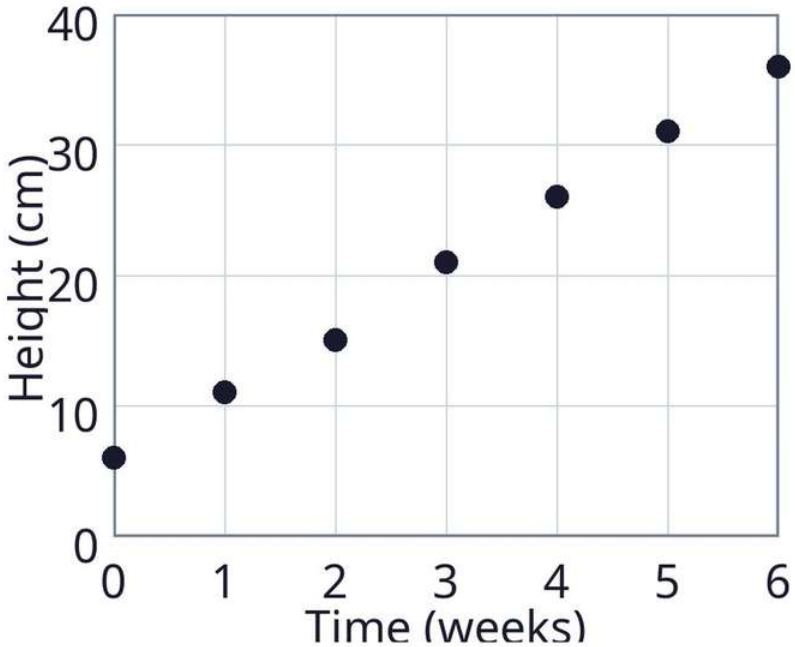
Question 13 · Scatter plots · Association

What is the association between the x- and y-values for this graph?



Question 14 · Scatter plots · Lines of fit

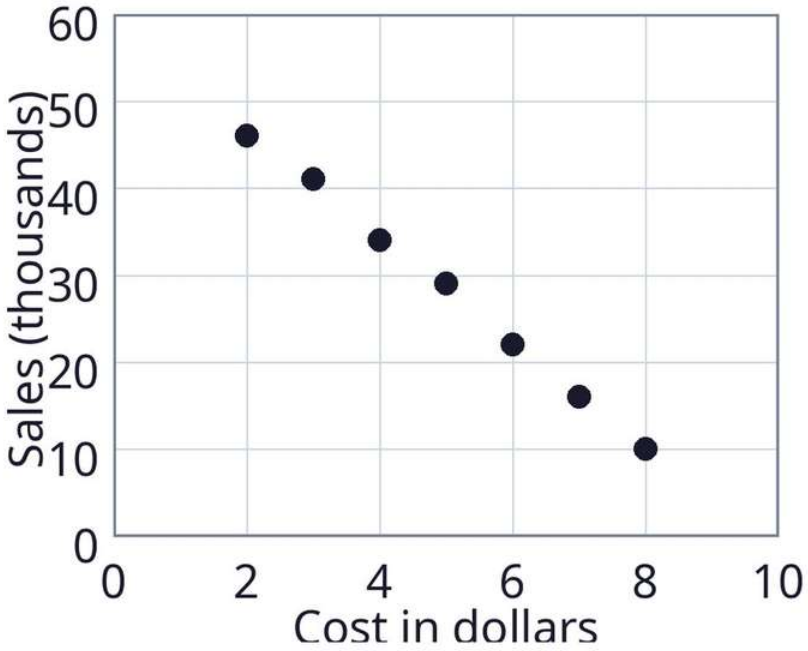
The scatter plot shows the height of a plant, in centimeters, measured each week.



Describe the type of correlation the scatter plot shows. Draw a trend line that models the data set on your writing pad, and find the equation of that trend line.

Question 15 · Scatter plots · Lines of fit

The scatter plot shows the weekly sales of a product, in thousands of units, at different prices.



Describe the type of correlation the scatter plot shows. Draw a trend line that models the data set on your writing pad, and find the equation of that trend line.

Question 16 · Scatter plots · Using a trend line

For the trend line in Question 14, what would the expected height of the plant be after 2 weeks? Explain what this means in the context of the data.

Question 17 · Scatter plots · Using a trend line

For the trend line in Question 15, what would the expected sales be if the cost was set at $6.50? Explain what this means in the context of the data.

Question 18 · Scatter plots · Reasoning

Would you expect the trend line for the plant's height to continue in the same direction indefinitely? Explain.